FACTORY REALLOCATION AND SHIPPING OPTIMIZATION RECOMMENDATION SYSTEM

A Data-Driven Decision-Support Study for Nassau Candy Distributor

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RESEARCH PAPER

Project focus: data validation, shipping lead-time analysis, predictive-model evaluation, and candidate factory-reallocation scenarios.

Report status: Analytical prototype. Results are preliminary and must be interpreted in light of the date-quality issues and model-validation outcomes described in this paper.

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Dataset analyzed: 10,194 order records

Abstract

This paper presents a decision-support prototype for analyzing shipping outcomes and exploring product reallocation among candidate factories for Nassau Candy Distributor. The study uses 10,194 order records and examines shipping modes, order and ship dates, product and regional attributes, sales, units, gross profit, and cost. The workflow combines data profiling, exploratory analysis, chronological model validation, and scenario-based factory-reallocation simulation. Initial analysis identifies a major data-quality concern: order dates span January 2024 to December 2025, while recorded ship dates span June 2026 to June 2030, producing calculated durations of approximately 904 to 1,642 days. These values require source-system verification before operational interpretation. Three regression models were compared; all achieved negative R^2 values on the reported validation split, indicating performance below a test-set mean-baseline benchmark. The prototype should therefore be treated as an exploratory decision-support tool, not as an automated source of reliable delivery forecasts or proven cost savings. The paper concludes with data remediation, validation, and operational-readiness recommendations.

Keywords: shipping lead time, factory reallocation, supply-chain analytics, regression, data quality, decision support

1. Introduction

Shipping performance influences customer experience, inventory planning, fulfillment costs, and service reliability. When an organization can consider multiple production or fulfillment locations, a structured comparison of candidate factories may help planners evaluate whether alternative allocations could improve delivery performance or profitability. Such decisions require reliable historical data, clear lead-time definitions, realistic capacity information, and defensible estimates of transportation costs.

The Nassau Candy Factory Reallocation and Shipping Optimization Recommendation System was developed as an analytical prototype. It combines historical order analysis, machine-learning experiments, and a simulation framework intended to compare current assignments with alternative factory candidates. A key principle of this work is that a functioning dashboard does not by itself establish that recommendations are correct. Data validity and out-of-sample performance must be assessed before operational use.

2. Problem Statement

The project asks how order, product, region, shipping-mode, and financial data can be organized into a decision-support workflow that helps analysts investigate shipping performance and alternative factory assignments. The initial dataset contains date patterns that generate exceptionally long calculated shipping durations. In addition, tested lead-time models do not outperform a simple mean-baseline benchmark on the reported chronological test set. These limitations prevent defensible claims about actual delivery improvement, savings, or factory feasibility. The immediate problem is therefore to establish a trustworthy analytical foundation and design a transparent simulation that can be validated when operational inputs are available.

3. Objectives

- Profile the historical dataset and document its coverage, variables, and data-quality risks.
- Explore shipping-mode composition and the available product, regional, and financial information.
- Compare candidate regression models for shipping lead-time prediction using chronological validation.
- Create a scenario framework to compare current factory assignments with alternative candidates.
- Identify limitations that affect the reliability and operational interpretation of results.
- Recommend practical data, modeling, and operational steps required before deployment.

4. Dataset Description

The primary file analyzed was *Nassau Candy Distributor.csv*, reported as containing 10,194 rows and 18 columns. Fields include record and order identifiers, order and ship dates, shipping mode, customer and geographic attributes, division and region, product identifiers and names, sales, units, gross profit, and cost. A feature-engineered Step 3 dataset was also created; a corrected version was generated after date-parsing issues were investigated.

Attribute	Description
Record count	10,194 historical order records
Order-date coverage	2 January 2024 to 31 December 2025
Recorded ship-date coverage	30 June 2026 to 28 June 2030
Shipping modes	Standard Class, Second Class, First Class, Same Day
Product and geography	15 products, 4 regions, and 59 states reported in project analysis
Financial and quantity fields	Sales, Units, Gross Profit, Cost
Primary quality concern	Date alignment and calculated shipping durations require source validation

4.1 Shipping-mode distribution

The counts below describe the distribution of records, not the relative speed or service quality of each mode. A valid mode comparison requires validated dates and an agreed definition of lead time.

Shipping mode	Records	Approx. share
Standard Class	6,120	60.0%
Second Class	1,979	19.4%
First Class	1,548	15.2%
Same Day	547	5.4%
Total	10,194	100.0%

4.2 Date-quality issue

The reported order dates run from January 2024 through December 2025, whereas ship dates run from June 2026 through June 2030. The calculated differences are reported as approximately 904 days at the minimum, 1,274 days at the median, 1,321 days on average, and 1,642 days at the maximum. These values are unusually long for ordinary order-to-shipment intervals. They may reflect synthetic dates, date-shift errors, an incorrect date definition, or a source-data issue; the available values alone do not establish which explanation is correct.

Interpretation rule: these durations are calculations from the available fields, not validated estimates of actual warehouse processing or transportation time. They should not be presented as confirmed operational shipping times.

5. Methodology

5.1 Data inspection and preparation

The workflow begins with row and column inspection, data-type checks, missing-value review, duplicate checks, and validation of key fields. Date parsing is particularly important because ambiguous day-month-year strings can be misinterpreted by software defaults. The project subsequently produced a corrected feature-engineered file and additional date-investigation scripts. Before modeling, the date format should be explicitly specified, and calculated durations should be checked for negative values, extreme values, and consistency with source-system records.

5.2 Exploratory data analysis

Exploratory analysis summarizes record counts, time coverage, shipping-mode mix, product and regional coverage, and the availability of financial measures. These summaries describe the available data and help reveal quality issues. Because the date anomaly remains important, any analysis relying on shipping lead time must be interpreted cautiously. Sales, units, gross profit, and cost can support commercial analysis only after their definitions, units, and reconciliation rules are confirmed.

5.3 Machine-learning evaluation

The project compared Gradient Boosting Regressor, Random Forest Regressor, and Linear Regression. The reported chronological split used 3,129 training records and 1,001 testing records. Training dates were reported as 2 January 2024 to 3 August 2025, and testing dates as 4 August 2025 to 12 December 2025. Chronological validation is more appropriate for time-dependent prediction than randomly mixing earlier and later observations. For reproducibility, the final report should also record the feature set, preprocessing steps, final filtering rules, and software versions.

5.4 Factory-reallocation simulation

The Step 9 simulation compares baseline assignments with alternative factory candidates across product and regional groupings. Five candidate factory names identified during the project are Lot's O' Nuts, Secret Factory, Sugar Shack, The Other Factory, and Wicked Choccy's. A candidate assignment is not automatically an optimized recommendation. Reliable optimization requires verified route distances and transport costs, actual factory capabilities and capacities, product-factory eligibility, production costs, and service constraints.

5.5 Decision-support presentation

The intended Streamlit dashboard includes product, region, and shipping-mode selectors; a speed-versus-profit priority control; a factory-optimization simulator; what-if analysis; a recommendation view; and a risk-and-impact panel. The dashboard should label estimates and assumptions, distinguish historical observations from simulated scenarios, and avoid presenting unvalidated predictions as guaranteed outcomes.

6. Exploratory Analysis and Key Observations

- **Dataset size:** the working dataset contains 10,194 records; this supports exploratory summaries but does not guarantee model quality.
- **Shipping-mode concentration:** Standard Class represents about 60% of records, while Same Day represents about 5.4%. This is descriptive, not evidence of service performance.
- **Temporal inconsistency:** the gap between order-date and ship-date ranges is the most important identified data-quality concern.
- **Financial and quantity fields:** sales, units, gross profit, and cost may support commercial analysis, subject to reconciliation and definition checks.
- **Geographic and factory mapping:** geographic fields can support regional analysis, but factory decisions require verified origins, destinations, and routing data.

7. Machine-Learning Evaluation

The model-comparison results recorded during Step 10 are shown below. MAE and RMSE are expressed in days. Lower MAE and RMSE are preferable; higher R^2 is generally preferable. Negative R^2 values indicate that the models performed worse than predicting the mean target value on the evaluated test set.

Model	MAE (days)	RMSE (days)	R^2
Gradient Boosting Regressor	176.33	182.50	-0.0089
Random Forest Regressor	177.24	186.03	-0.0484

Model	MAE (days)	RMSE (days)	R^2
Linear Regression	180.36	183.05	-0.0150

Gradient Boosting Regressor has the lowest MAE and RMSE among the three reported results, but its R^2 remains negative. Differences in error metrics are small relative to the overall errors. It would be inappropriate to declare this model production-ready or to treat the results as evidence of reliable predictions of actual shipping times. Data-quality issues may contribute to weak performance, but that relationship must be tested rather than assumed.

7.1 Evaluation implications

- Do not use current model outputs to promise delivery dates or service-level improvements.
- Rebuild the target only after validating the meaning and correctness of the date fields.
- Compare models against a simple baseline on a strictly chronological holdout.
- Ensure preprocessing is fitted only on training data and inspect residual diagnostics.
- Evaluate performance by mode, region, product, and time period where sample sizes are adequate.

8. Factory-Reallocation Simulation

The project simulation output was reported as containing 265 scenario rows: 53 baseline product-region rows and 212 alternative candidates, or four alternatives per baseline row. This structure supports side-by-side scenario inspection. It does not establish that all alternatives are feasible, that the best alternative has been selected, or that moving production would improve profit or delivery performance.

8.1 Required decision variables

- Product-to-factory compatibility and production capability.
- Available capacity, utilization, labor, and production lead time by factory.
- Origin-destination distance, freight rates, mode availability, and handling costs.
- Validated historical lead time or a calibrated estimate for each feasible route.
- Unit economics, demand, gross profit, service commitments, and late-delivery penalties.
- Constraints such as minimum order quantities, geographic restrictions, and customer requirements.

8.2 Suggested objective

A future optimization formulation could minimize a weighted combination of expected fulfillment cost and late-delivery risk, subject to capacity and eligibility constraints. A speed-versus-profit control can change the displayed weighting, but the weights and trade-offs should be visible. The objective should not be treated as validated until cost, capacity, and service inputs are measured and checked.

9. Findings

- The project has assembled an order-level dataset and an analytical workflow for exploratory analysis, modeling, and scenario generation.
- Standard Class is the most common mode, accounting for approximately 60% of records.
- The order and ship date ranges generate extremely long calculated durations and require source-system investigation.
- All three tested regression models have negative R^2 values on the reported test split, so predictive performance is not yet adequate for operational use.

- The factory simulation organizes baseline and alternative scenarios, but scenario counts do not prove feasibility, optimality, or financial benefit.
- The dashboard can support analyst exploration if it communicates limitations, assumptions, and uncertainty clearly.

10. Limitations

- **Data validity:** date fields may not represent true order-to-shipment timing; the cause is not confirmed.
- **Model quality:** reported R^2 values are negative and test errors remain substantial.
- **Operational inputs:** verified factory capacity, product eligibility, route-level freight cost, and production constraints are not established in the reported results.
- **Financial claims:** there is no validated savings estimate or causal comparison from the current scenario structure.
- **Reproducibility:** feature definitions, preprocessing, random seeds, software versions, and filtering rules should be documented.
- **External validity:** results may not generalize to future demand, other routes, or changed operating conditions.

11. Recommendations

11.1 Immediate priorities

- Confirm the date format and field meanings with the data owner and compare sampled records against source-system evidence.
- Document whether ship date means dispatch, delivery, a synthetic date, or another event.
- Recalculate lead time only after the target definition is agreed; quarantine unresolved records rather than silently correcting them.
- Automate checks for missing dates, impossible chronology, extreme durations, duplicate identifiers, and financial reconciliation.
- Show visible dashboard warnings that current lead-time predictions and savings estimates are not validated.

11.2 Modeling and simulation improvements

- Retrain models after data remediation and compare them with a baseline using a chronological holdout.
- Inspect residuals and performance by product, region, and shipping mode; determine whether a simpler model is sufficient.
- Add route-level transportation cost and factory-capacity data before estimating financial impact.
- Validate recommendations with operations stakeholders and test them on historical or pilot scenarios.
- Report uncertainty ranges and sensitivity analyses rather than presenting single-point estimates as guarantees.

11.3 Deployment governance

- Treat the dashboard as an analyst decision-support tool until the model passes agreed acceptance thresholds.
- Record dataset and model versions, assumptions, constraints, and the rationale behind each recommendation.
- Require human review before changing production assignments or customer commitments.
- Monitor data drift, prediction errors, realized delivery performance, and realized versus predicted cost after a pilot.

12. Conclusion

This study establishes an initial framework for examining shipping records, testing lead-time prediction models, and exploring factory-reallocation scenarios for Nassau Candy Distributor. Its main contribution is a structured analytical workflow and a transparent view of the evidence required for responsible optimization. The findings also show why operational deployment would be premature: calculated shipping durations are unusually long, and all evaluated models underperform the mean-baseline benchmark on the reported test set. The next step is to validate source data, rebuild and re-evaluate the target and models, and add real capacity and cost constraints. With those foundations in place, the Streamlit dashboard can evolve into a more credible decision-support system that helps stakeholders compare trade-offs while retaining human oversight.

Appendix A. Reported Project Artifacts

Artifact	Purpose
Nassau Candy Distributor.csv	Primary historical dataset used for initial analysis
Nassau_Candy_Feature_Engineered_Step3_CORRECTED.csv	Corrected feature-engineered dataset produced during date investigation
step10_time_based_validation.py	Chronological model-validation workflow
step11_data_quality_investigation.py	Data-quality investigation
step12_root_cause_analysis.py	Follow-up analysis of potential date issues
step13_date_pattern_investigation.py	Date-pattern investigation
step9_outputs/factory_reallocation_simulation.csv	Scenario table for baseline and alternative factory candidates
app.py	Streamlit dashboard entry point

Appendix B. Interpretation Notes

Numerical values in this paper are based on project outputs reported during development. Where source-level validation has not been completed, values are labeled as reported or calculated rather than confirmed operational facts. Candidate factory names and simulation rows identify scenario inputs; they do not certify factory capability or recommend a production transfer. Before publication or external stakeholder use, reproduce the calculations from the final committed dataset and attach a versioned data dictionary and model-evaluation output.